

Supplementary Information: Data Quality Assurance Analysis Keheala Study 2

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This supplementary document presents descriptive statistics and data quality assurance (DQA) analyses conducted as part of the Keheala Study 2 randomized controlled trial, which evaluated a digital adherence support platform for tuberculosis (TB) patients in Kenya. The DQA assesses recording errors in TIBU, Kenya’s national electronic TB reporting system, by treating a contemporaneous paper registry audit as the ground truth. The TIBU dataset available for this study covers patients registered between January 2017 and December 2021 ($N = 119,653$ nationwide). A paper registry audit was conducted across study clinics between November 2018 and September 2019, yielding 3,645 audited records, of which 3,370 could be matched to a corresponding TIBU record and form the basis of the DQA. Tables 1a and 1b characterize the study population across three nested cohorts: all patients registered in TIBU during the study period, patients at study clinics in TIBU (the MITT sample), and the subset of study-clinic patients whose records were audited in the DQA.

The core of the DQA compares treatment outcomes recorded in TIBU against outcomes recorded in the paper registry for each matched patient. Three types of discrepancy are distinguished. Type I errors occur when TIBU records a successful outcome (treatment completion or cure) but the paper registry records a failure (death, treatment failure, or loss to follow-up), representing a false-positive in the digital record. Type II errors are the reverse: TIBU records a failure while the paper registry records success, representing a false-negative. Incomplete-data discrepancies arise when TIBU contains no outcome but the paper registry does. The cross-tabulation in Table 2 maps the full distribution of these discrepancies across outcome categories; subsequent tables stratify error rates by clinic characteristics (province, urbanicity, and size) and patient characteristics (sex, age, disease type, and HIV status); and the time-series figures trace how error rates vary with the age of the patient record and across the study period.

1 Patient Characteristics

Tables 1a and 1b describe the study population across three cohorts: all patients registered in the TIBU electronic system during the study period, patients at study clinics recorded in TIBU (MITT sample), and patients at study clinics whose records were audited in the paper registry DQA.

Table 1a: Individual and disease characteristics across TIBU and paper registry cohorts.

Characteristic	Digital				Paper	
	All		Study clinics		Study clinics	
	(N=119,653)		(N=15,487)		(N=3,370)	
	N	(%)	N	(%)	N	(%)
<i>Individual characteristics</i>						
Male	75,597	(63.2%)	9,850	(63.6%)	2,104	(64.5%)
Mean age (years)	36.1		34.4		32.7	
<i>Disease characteristics</i>						
Bacteriologically confirmed	62,149	(51.9%)	11,663	(75.3%)	2,295	(70.4%)
Drug resistant	1,343	(1.1%)	2	(0.0%)	3	(0.1%)
Extrapulmonary TB	17,558	(14.7%)	1,834	(11.8%)	369	(11.3%)
HIV coinfection	32,077	(26.8%)	4,118	(26.8%)	746	(23.3%)
Retreatment	8,351	(7.0%)	1,308	(8.7%)	297	(9.3%)

Table 1b: Treatment outcomes across TIBU and paper registry cohorts.

Characteristic	Digital				Paper	
	All		Study clinics		Study clinics	
	(N=119,653)		(N=15,487)		(N=3,370)	
	N	(%)	N	(%)	N	(%)
<i>Outcomes</i>						
Cured	25,399	(21.2%)	7,008	(45.3%)	1,148	(34.1%)
Treatment completed	31,719	(26.5%)	6,851	(44.2%)	1,520	(45.1%)
Died	4,911	(4.1%)	787	(5.1%)	186	(5.5%)
Treatment failed	382	(0.3%)	129	(0.8%)	37	(1.1%)
Lost to follow-up	3,594	(3.0%)	712	(4.6%)	133	(3.9%)
Not completed	47,184	(39.4%)	0	(0.0%)	87	(2.6%)
Transfer out	6,382	(5.3%)	0	(0.0%)	127	(3.8%)
MT4	0	(0.0%)	0	(0.0%)	12	(0.4%)
Not TB	0	(0.0%)	0	(0.0%)	0	(0.0%)
Missing	82	(0.1%)	0	(0.0%)	120	(3.6%)

2 Data Quality Assurance: Outcome Discrepancies

2.1 Crosstab of TIBU vs. Paper Registry Outcomes

Table 2 shows the full cross-tabulation of treatment outcomes recorded in TIBU against outcomes recorded in the paper registry for all matched patients ($N = 3,370$).

Table 2: Cross-tabulation of treatment outcomes in TIBU vs. paper registry.

Outcome in TIBU	Treatment Outcome in Paper Registry							Total
	Blank	C	D	F	LTFU	TC		
	N (%)	N (%)	N (%)	N (%)	N (%)	N (%)		
Blank	80 (66.7%)	22 (18.3%)	2 (1.7%)	2 (1.7%)	2 (1.7%)	12 (10.0%)		120
C	45 (3.9%)	1,032 (89.9%)	1 (0.1%)	3 (0.3%)	7 (0.6%)	60 (5.2%)		1,148
D	4 (2.2%)	3 (1.6%)	170 (91.4%)	0 (0.0%)	1 (0.5%)	8 (4.3%)		186
F	0 (0.0%)	1 (2.7%)	1 (2.7%)	28 (75.7%)	5 (13.5%)	2 (5.4%)		37
LTFU	3 (2.3%)	2 (1.5%)	1 (0.8%)	0 (0.0%)	121 (91.0%)	6 (4.5%)		133
MT4	2 (16.7%)	3 (25.0%)	0 (0.0%)	4 (33.3%)	1 (8.3%)	1 (8.3%)		12
NC	10 (11.5%)	34 (39.1%)	6 (6.9%)	0 (0.0%)	4 (4.6%)	32 (36.8%)		87
TC	34 (2.2%)	81 (5.3%)	8 (0.5%)	0 (0.0%)	13 (0.9%)	1,384 (91.1%)		1,520
TO	4 (3.1%)	43 (33.9%)	5 (3.9%)	0 (0.0%)	13 (10.2%)	62 (48.8%)		127
Total	182	1,221	194	37	167	1,567		3,370

Note: Percentages are row shares.

Red = Type I error (TIBU: success; paper: failure).

Green = Type II error (TIBU: failure; paper: success).

Blue = Incomplete data (TIBU: no outcome; paper: outcome present).

2.2 Error Rates by Clinic Characteristics

Table 3 reports false-negative, false-positive, and false-missing rates stratified by province, urbanicity, and clinic size.

Table 3: DQA error rates by province, urbanicity, and clinic size.

	<i>N</i>	(%)	False positive <i>N</i>	(%)	False negative <i>N</i>	(%)	Incomplete data <i>N</i>	(%)
<i>By province</i>								
Central	19	(0.6%)	4	(21.1%)	0	(0.0%)	1	(5.3%)
Coast	745	(23.9%)	14	(1.9%)	7	(0.9%)	32	(4.3%)
Eastern North	2	(0.1%)	0	(0.0%)	0	(0.0%)	0	(0.0%)
Eastern South	287	(9.2%)	22	(7.7%)	4	(1.4%)	23	(8.0%)
Nairobi	378	(12.2%)	8	(2.1%)	6	(1.6%)	2	(0.5%)
North Eastern	748	(24.0%)	5	(0.7%)	9	(1.2%)	2	(0.3%)
Nyanza North	394	(12.7%)	9	(2.3%)	2	(0.5%)	4	(1.0%)
Nyanza South	43	(1.4%)	0	(0.0%)	1	(2.3%)	0	(0.0%)
Rift Valley North	44	(1.4%)	0	(0.0%)	0	(0.0%)	3	(6.8%)
Rift Valley South	419	(13.5%)	25	(6.0%)	2	(0.5%)	28	(6.7%)
Western	32	(1.0%)	1	(3.1%)	1	(3.1%)	1	(3.1%)
<i>By urban/rural</i>								
Urban	985	(31.7%)	22	(2.2%)	11	(1.1%)	21	(2.1%)
Rural	2,122	(68.2%)	66	(3.1%)	21	(1.0%)	75	(3.5%)
<i>By clinic size</i>								
Large clinic	2,438	(78.4%)	58	(2.4%)	30	(1.2%)	84	(3.4%)
Small clinic	673	(21.6%)	30	(4.5%)	2	(0.3%)	12	(1.8%)
All	3,111	(100.0%)	88	(2.8%)	32	(1.0%)	96	(3.1%)

Note: Urban/rural classification is based on clinic records. Large clinics are those with at least 2 patients registered in TIBU (median); small clinics are those with fewer.

2.3 Error Rates by Patient Characteristics

Table 4 reports false-negative, false-positive, and false-missing rates broken down by individual characteristics and disease characteristics.

Table 4: DQA error rates by individual and disease characteristics.

	<i>N</i> (%)		False positive <i>N</i> (%)		False negative <i>N</i> (%)		Incomplete data <i>N</i> (%)	
<i>Individual characteristics</i>								
Male (%)	1,992	(64.0%)	70	(3.5%)	20	(1.0%)	58	(2.9%)
<i>Age group</i>								
<15	438	(14.1%)	6	(1.4%)	1	(0.2%)	6	(1.4%)
15–34	1,418	(45.6%)	41	(2.9%)	18	(1.3%)	43	(3.0%)
35–54	929	(29.9%)	28	(3.0%)	7	(0.8%)	34	(3.7%)
55+	315	(10.1%)	8	(2.5%)	6	(1.9%)	10	(3.2%)
<i>Disease characteristics</i>								
Bacteriologically confirmed	2,177	(70.0%)	62	(2.8%)	26	(1.2%)	75	(3.4%)
Not bacteriologically confirmed	934	(30.0%)	26	(2.8%)	6	(0.6%)	21	(2.2%)
Extrapulmonary	351	(11.3%)	7	(2.0%)	2	(0.6%)	13	(3.7%)
Pulmonary	2,754	(88.5%)	81	(2.9%)	30	(1.1%)	80	(2.9%)
Retreatment	290	(9.3%)	5	(1.7%)	5	(1.7%)	9	(3.1%)
New case	2,755	(88.6%)	81	(2.9%)	26	(0.9%)	67	(2.4%)
HIV positive	720	(23.1%)	23	(3.2%)	7	(1.0%)	29	(4.0%)
HIV negative	2,330	(74.9%)	62	(2.7%)	24	(1.0%)	66	(2.8%)
All	3,111	(100.0%)	88	(2.8%)	32	(1.0%)	96	(3.1%)

2.4 Age of Record by Error Type

Table 5 reports the age of record (days from date of registration to TIBU outcome date) stratified by error type.

Table 5: Age of record by error category (in days from registration to TIBU outcome date).

Error type	N	(%)	Mean	Median
False positive (Type I)	30	(1.0%)	163	182
False negative (Type II)	17	(0.6%)	100	103
False missing	81	(2.8%)	157	148
No error	2,802	(95.6%)	190	170

Note: Age of record is the number of days that have passed since the patient’s registration date to the treatment outcome date recorded in TIBU. Records with negative values excluded.

2.5 Error Rates by Age of Record

Figure 2 plots DQA error rates by age of record (days from registration to TIBU outcome date), binned in 90-day (quarterly) intervals with a 3-quarter centered rolling mean ($N \geq 10$ per bin). Figure 3 restricts to bins with $N \geq 100$ to reduce noise from small samples.

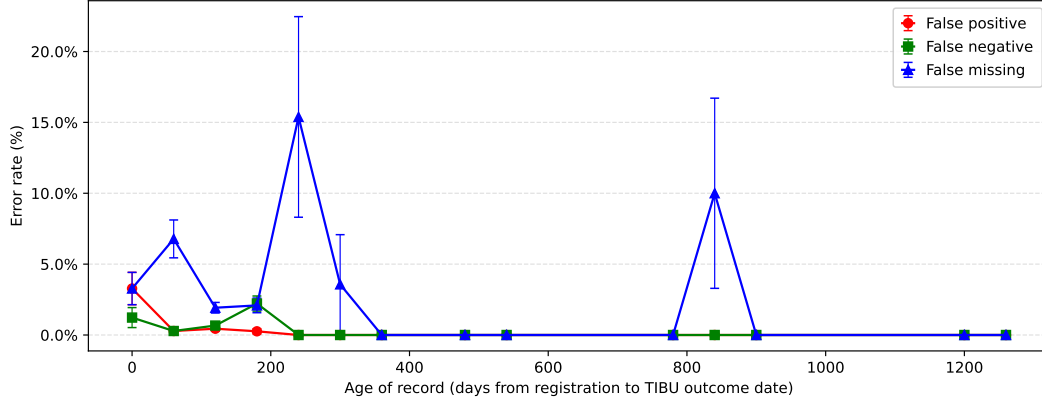


Figure 2: DQA error rates by age of record (bins with $N \geq 10$).

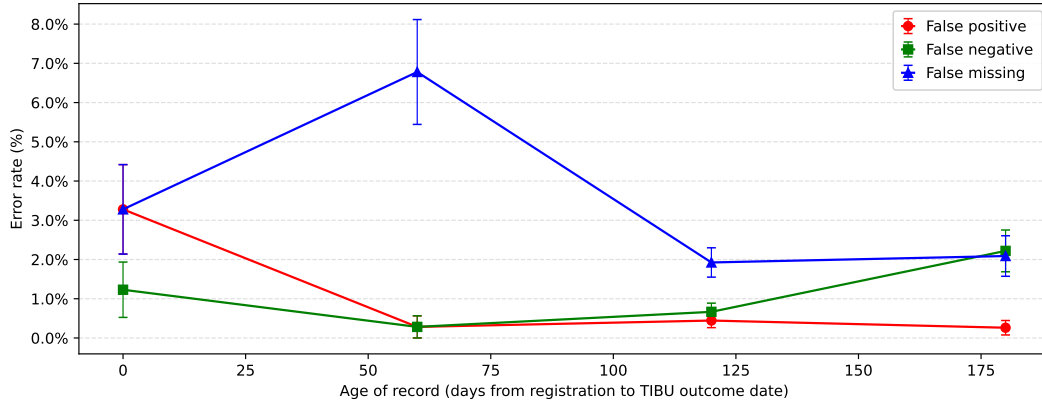


Figure 3: DQA error rates by age of record, restricted to bins with $N \geq 100$.

2.6 Error Rates Over Time

Figure 4 plots DQA error rates across the study period, binned by quarter with a 3-quarter centered rolling mean ($N \geq 10$ per quarter). Figure 5 restricts to quarters with $N \geq 100$ to reduce noise from small samples.

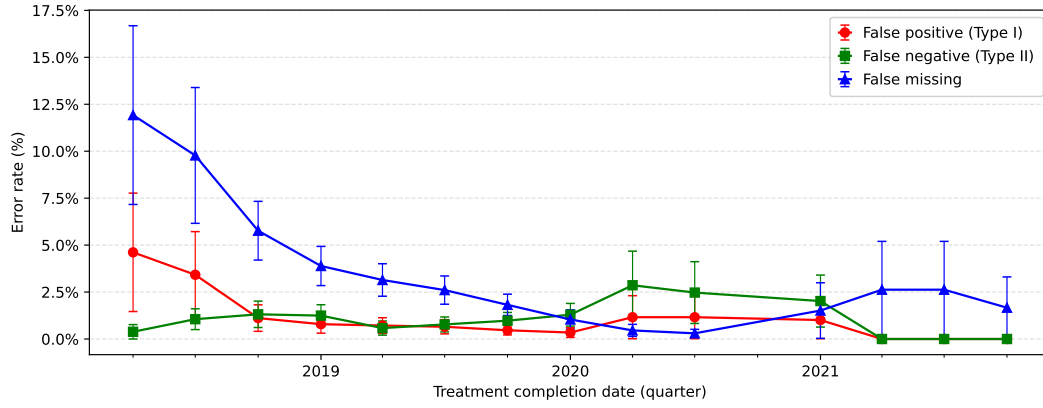


Figure 4: DQA error rates over time, binned by quarter with a 3-quarter centered rolling mean ($N \geq 10$ per quarter).

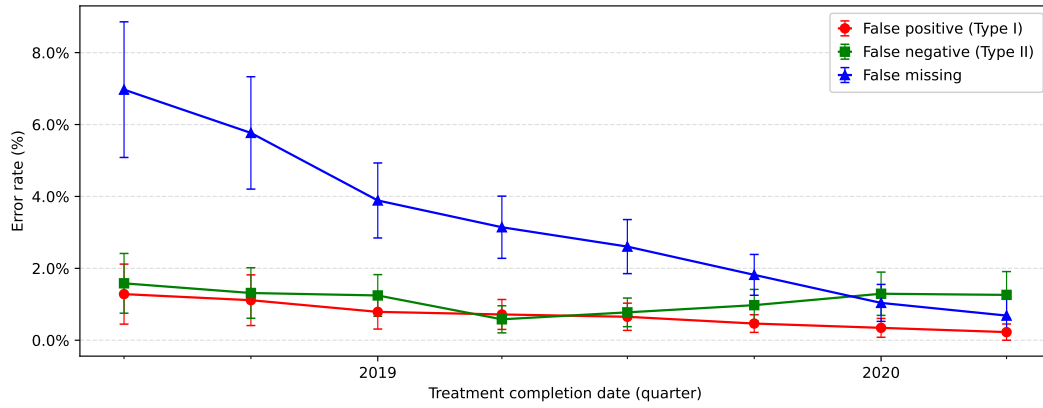


Figure 5: DQA error rates over time, binned by quarter with a 3-quarter centered rolling mean ($N \geq 100$ per quarter).